

PARTS LIST

Semiconductors:

IC1	- 78L05, 5V regulator
IC2, IC5	- LM358 dual operational amplifier
IC3	- LM339 quad comparator
IC4	- TL494 pulse-width modulation control chip
IC6	- LM319 dual comparator
IC7	- LM35DZ centigrade temperature sensor
T1, T2	- IRF530 p-channel power MOSFET
T3, T4	- 2N7000 n-channel enhancement MOSFET
T5	- BC548 npn transistor
T6, T7	- TIP120 npn Darlington transistor
T8	- TIP125 pnp Darlington transistor
D1, D2	- MBR2045 Schottky rectifier diode
D3	- 1N4007 rectifier diode
D4, D5	- 1N4148 signal diode
LED1-LED3	- 5mm LED
MOV1	- 14D431 K varistor
PVM1	- PM80 80 watt solar module

Resistors (all 1/4-watt, 5% carbon unless stated otherwise):

R1, R2, R13, R22-1M	
R3, R4, R15	
R29, R30, R31, R41, R42, R43, R44	
R5, R6	- 100-kilo-ohm
R7, R23, R25	- 0.1-ohm, 5W
R8, R9, R14	- 2.2-kilo-ohm
R10	- 1-kilo-ohm
R12, R16, R17, R19, R20, R24, R27, R34, R50	- 8.2-kilo-ohm
R18, R32, R46	- 10-kilo-ohm
R39	- 47-kilo-ohm
R21	- 51-kilo-ohm
R26	- 27-kilo-ohm
R28	- 22-kilo-ohm
R33	- 470-kilo-ohm
R35	- 20-kilo-ohm
R36	- 33-kilo-ohm
R37, R38	- 18-kilo-ohm
R40	- 510-ohm
R45	- 5-kilo-ohm
R47	- 15-kilo-ohm
R48	- 1.5-kilo-ohm
R49	- 75-ohm
R51	- 1.2-kilo-ohm
R52	- 4.7-kilo-ohm
R53	- 5.6-kilo-ohm
VR1	- 1-kilo-ohm, preset
VR2	- 100-kilo-ohm, preset

Capacitors:

C1, C2, C4, C5, C6, C9	
C3	- 0.1 μ F ceramic
C7, C8	- 100 μ F, 25V electrolytic
C10	- 10 μ F, 25V electrolytic
C11	- 0.01 μ F ceramic
C12	- 2.2 μ F, 16V electrolytic
C12	- 1 μ F, 16V electrolytic

Miscellaneous:

Batt.	- 12V, 80Ah battery
F1	- 10A Fuse
S1	- On/off switch
S2, S3, S4	- Tactile switch
Coaxial cable	- 75-ohm
	- jumper connectors
	- Berg strips

of 17.5V-18.5V will provide improved MPPT and battery charging performance. For 24V or 48V batteries, these voltages should be multiplied by 2 and

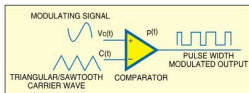


Fig. 3: Basic conventional PWM circuit

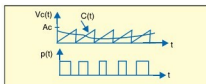


Fig. 4: Variable duty ratio generation using PWM

4, respectively.

Minimum acceptable I_{mp} current rating. A preferred rule of thumb is that there should be at least 3 amperes of I_{mp} per 100 amp-hours of battery capacity for proper battery charging. Thus it will be safe to select a PV module of 70W-80W P_{max} rating to get an I_{mp} of 4 to 4.5 amperes.

PV solar panel orientation

The parameters of a PV panel are specified by the manufacturer at 20°C cell temperature for the condition that each one square metre of its area receives 'peak sun' power of 1000W, thus providing 1000 Wh per m^2 (1 kWh/ m^2)—representing the solar energy received in one hour on a cloudless summer day when the PV panel is directed towards the sun. In practice, the power output of a photovoltaic solar panel is affected by its orientation with respect to the sun and atmospheric conditions.

The best tilt angle is usually equal to the latitude of the installation site. For example, New Delhi is at latitude 28°-36' North and if solar panels are tilted at 28°-36' (with respect to the horizon), these would be perpendicular to the sun twice a year. As the Indian sub-continent is towards north of the equator, the PV panel should face towards south in the azimuth. When you have a large number of PV panels, some of them may be tilted within a range of $\pm 15^\circ$ of the site-latitude, depending on whether a slight winter/

summer bias is desirable in the system.

Insolation. This term is used for sunlight intensity in relation to the solar PV panel and is measured in equivalent full or peak sun hours (as explained above) during the day (24 hours). Even if the sun is in the horizon for a period exceeding twelve hours in a day, your PV panel may receive full sunlight only for four to seven hours or so. This is due to the fact that the panel will not be perpendicular to the sunrays most of the time and also there is earth atmospheric absorption.

Most of the Indian subcontinent receives full sunlight for about five hours a day. (Refer to the map at www.solar4power.com/map14-global-solar-power.html). Taking the conversion efficiency of the system into account, a PV panel rated for P_{max} of 75 watts will be able to supply you 75 watts of power for just five hours per day. Thus you would be able to draw only 75W $\times$ 5 hours = 375 watt-hours of energy per day from the PV panel.

If you desire to run your appliance for a longer duration, you will have to install more number of PV panels and connect them in series/parallel for increasing the combination's voltage/current rating.

In India, various companies like Central Electronics Ltd (CEL), Bharat Electronics Ltd (BEL), Bharat Heavy Electricals Ltd (BHEL), Moser Baer Photovoltaic, TATA-BP Solar and Mahatma Solar Technology are making solar panels using indigenous and imported technologies. A PV module such as the PM80 from CEL, Sahibabad, having the specifications as shown in the table on previous page, could be used for our example case.

Battery selection

The selected PV module should be able to support an energy supply of 375 watt-hours per day. That means if you have an appliance rated at 40 watts which is supplied power from an